

upGrad

Foundations: Python, Cloud and a reproducible workspace

M00 Session 01

Session map

01 Set the workspace boundary

Separate the project, credentials, data and documentation from the start.

02 Make the workflow explicit

Use an environment, version control and clear responsibility boundaries.

03 Apply the boundary

Use a concise exit prompt to check the learner's mental model.

Each topic can be compiled alone while this session file assembles the complete learner deck.

TOPIC 01

Set the workspace boundary

Separate the project, credentials, data and documentation from the start.

What this topic establishes

By the end of this topic, learners can

- describe the role of Python, IAM, S3 and Git in a data-science workflow;
- create a minimal local project structure that can be reproduced by a teammate; and
- distinguish credentials, code and data artefacts by where they should live.

The smallest useful data-science workspace

Local project

- versioned Python environment
- `src/` for reusable code
- `notebooks/` for exploration
- `README.md` for the next learner

AWS account boundary

- IAM role grants least-privilege access
- S3 stores durable data and artefacts
- CloudShell offers a browser-based terminal
- never commit keys or tokens to Git

TOPIC 02

Make the workflow explicit

Use an environment, version control and clear responsibility boundaries.

First checkpoint: prove the environment is explicit

```
1 python --version
2 python -m venv .venv
3 source .venv/bin/activate
4 python -m pip install pandas scikit-learn
5 git init
6 git status
7
```

Watch for – Do not treat an activated environment as documentation

Record dependencies in a requirements file or equivalent lockfile so the session can be reproduced later.

A practical responsibility map



Key takeaway – Reproducibility begins before modelling

A clear project boundary makes it possible to tell which code, data and credentials created an analysis.

TOPIC 03

Apply the boundary

Use a concise exit prompt to check the learner's mental model.

Exit prompt

Try it – Two minutes

Sketch a folder structure for a notebook that reads a CSV from S3 and trains a baseline model. Which files belong in Git, and which should be excluded?